

ARPAN SAHA

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SUMMARY

An M.Sc. Mathematics graduate from Indian Institute of Technology Madras specializing in the analysis and numerical methods of partial differential equations. Possesses proven research experience in modeling complex biological and physical systems, bridging probability theory and PDEs, utilizing the Finite Element Method to complement abstract qualitative theory with robust numerical insights.

EDUCATION

- Indian Institute of Technology Madras | Chennai, Tamil Nadu, India** **2024 - 2026**
Master of Science in Mathematics | CGPA: 8.41
- Ramakrishna Mission Vidyamandira | Belur, West Bengal, India** **2021 - 2024**
Bachelor of Science (Honours) in Mathematics | CGPA: 9.22

RELEVANT COURSEWORK

- Core Courses:** Real & complex analysis, ordinary & partial differential equations, multivariable calculus, numerical analysis, abstract & linear algebra, metric spaces, probability theory & statistics, measure and integration, topology, combinatorics, functional analysis.
- Elective Courses:** advanced differential equations, applied integral equations, Fourier analysis, numerical linear algebra, Sobolev spaces and PDEs, stochastic calculus for finance.

RESEARCH INTERESTS

Currently, my research interests lie in analysis and numerical methods for partial differential equations, focusing on finite element methods.

ACADEMIC PROJECTS

- Indian Institute of Technology Madras** **Jan 2026 - May 2026**
Master's Project | Supervisor: Dr. Bidhan Chandra Sardar *Chennai, Tamil Nadu*
 - Conducted a detailed study on the weak formulation of second-order elliptic PDEs on mixed boundary conditions from research literature.
 - Implemented finite element method for dirichlet boundary conditions using MATLAB, followed by error estimates and convergence analysis.
 - Reviewed the convergence behaviour depending on mesh size and domain geometry.
- Indian Institute of Technology Madras** **Aug 2025 - Nov 2025**
Postgraduate Seminar | Supervisor: Dr. Bidhan Chandra Sardar *Chennai, Tamil Nadu*
 - Conducted a detailed study on distribution theory and an introduction to Sobolev spaces.
- Ramakrishna Mission Vidyamandira** **Jan 2024 - May 2024**
Undergraduate Dissertation | Supervisor: Dr. Saugata Mitra *Belur, West Bengal*
 - Conducted a detailed study on dynamical systems focusing on its applications on cosmological models.
 - Reviewed research literature on cosmology, and reproduced the key analytical results and simulations using Maple.

TRAINING AND RESEARCH EXPERIENCE

- **Indian Institute of Technology Kanpur** **May 2025 - July 2025**
Kanpur, Uttar Pradesh
Students Undergraduate Research Graduate Excellence (SURGE) Program 2025
Supervisor: Prof. Malay Banerjee
 - Reviewed research literature regarding the Predator-Prey dynamics, and its relations with several natural factors.
 - Reproduced the key analytical results and visualized the behaviour of the systems by MATLAB simulations.
- **National Institute of Science Education and Research Bhubaneswar** **May 2024 - July 2024**
Jatni, Odisha
Summer Intern | Supervisor: Dr. Anupam Pal Choudhury
 - Worked on a reading project on the connection between random walk and heat equation.
- **Ramakrishna Mission Vivekananda Educational and Research Institute** **June 2023 - August 2023**
Belur, West Bengal
Summer Intern | Supervisor: Dr. Debaldev Jana
 - Studied the fundamentals of dynamical systems focusing on stability and bifurcation analysis.
 - Reproduced the key analytical results and numerical simulations using MATLAB from a research paper.
- **Indian Institute of Technology (Indian School of Mines) Dhanbad** **May 2023 - June 2023**
Dhanbad, Jharkhand
Mathematics Training and Talent Search Programme 2023
 - Attended lectures on real analysis, abstract & linear algebra, mathematical logic.
 - Actively participated in student seminars, group discussions.

PRESENTS & TALKS

- Presented a topic entitled “**Distribution Theory and Sobolev Spaces**” under course curriculum as a part of evaluation of postgraduate seminar at Indian Institute of Technology Madras. (November 24, 2025)
- Presented a poster entitled, “**Spatial Redistribution of Predator-Prey Dynamics**” as a part of Students Undergraduate Research Graduate Excellence Program 2025, organized by Office of Outreach Activities, Indian Institute of Technology Kanpur. (July 9, 2025)
- Presented a topic entitled, “**Dynamical Systems: Theory to Applications**” under course curriculum as a part of evaluation of B.Sc. dissertation at Ramakrishna Mission Vidyamandira. (May 18, 2024)
- Delivered a talk on “**Bolzano Weierstrass Theorem (for Sequences)**” in a student seminar on Mathematics Training and Talent Search Programme at IIT(ISM) Dhanbad. (June 21, 2023)

SEMINARS & WORKSHOPS

- Attended a seminar entitled “**From Poincaré to Saint Venant via Donati, Lions and Korn**”, organized by Department of Mathematics, Indian Institute of Technology Madras.
- Attended a workshop on “**Differential Equations through the Lens of Dynamical Systems**”, organized by Department of Mathematics, Ramakrishna Mission Vidyamandira. (April 2023)
- Attended a seminar entitled “**Importance of Abstract Algebra and Number Theory in Digital World**”, organized by Department of Mathematics, Ramakrishna Mission Residential College, Narendrapur. (April 28, 2023)

ACADEMIC ACHIEVEMENTS

- Qualified GATE examination in Mathematics with All India Rank 167. (2025)
- Qualified IIT JAM examination in Mathematics with All India Rank 186. (2024)
- Selected in Early Admit MSc at IIT Gandhinagar based on academic performance and interview. (2024)
- Received Radharani Barman Memorial Prize for academic excellence in undergraduate first year. (2022)
- Received INSPIRE Scholarship under the Scholarship for Higher Education (SHE) Programme for being in the top 1% students in 12th board and continuing till date. (2021)

TECHNICAL SKILLS

- **Mathematical Software:** MATLAB, Maple.
- **Programming Languages:** C, Python.
- **Typesetting & Document Preparation:** L^AT_EX, MS Word, MS Power Point.